

Nuclear Physics Angular Momentum Coupling Guide

This document provides the theoretical background and practical usage guide for the `angular_momentum_coupling.py` utility. This tool calculates allowed final nuclear states based on quantum mechanical selection rules.

1. Theoretical Basis

The code implements the **Channel Spin Coupling Scheme**, which is a standard method for determining angular momentum conservation in nuclear reactions.

Conservation Laws

In any nuclear reaction $\text{Target} + \text{Particle} \rightarrow \text{Final_State}$, the following quantities are conserved:

1. **Total Angular Momentum** ($\vec{J}$):

$$\vec{J}_{\text{final}} = \vec{J}_{\text{target}} + \vec{s}_{\text{particle}} + \vec{\ell}$$

2. **Parity** (π):

$$\pi_{\text{final}} = \pi_{\text{target}} \times \pi_{\text{particle}} \times (-1)^{\ell}$$

Where:

- $\vec{J}_{\text{target}}$: Spin of the target nucleus.
- $\vec{s}_{\text{particle}}$: Intrinsic spin of the interacting particle (projectile or transferred cluster).
- $\vec{\ell}$: Relative orbital angular momentum between the target and the particle.

Coupling Scheme Used in Code

The code performs the vector addition in two steps (Channel Spin representation):

1. **Calculate Channel Spin** ($\vec{S}$): Couples the target spin and particle spin.

$$\vec{S} = \vec{J}_{\text{target}} + \vec{s}_{\text{particle}}$$

Possible values: $|J_{\text{target}} - s_{\text{particle}}| \leq S \leq J_{\text{target}} + s_{\text{particle}}$

2. **Calculate Final Spin** ($\vec{J}_{\text{final}}$): Couples the channel spin with the orbital angular momentum.

$$\vec{J}_{\text{final}} = \vec{S} + \vec{\ell}$$

Possible values: $|S - \ell| \leq J_{\text{final}} \leq S + \ell$

2. Application to Different Reaction Types

To use the code correctly, you must identify the correct "Particle" input based on the reaction mechanism.

A. Resonance and Capture Reactions

Examples: (p, γ) , (n, γ) , (p, p')

In these reactions, a projectile fuses with the target to form a compound nucleus (resonance).

- **Target Input:** Spin/Parity of the target nucleus.
- **Particle Input:** Spin/Parity of the **projectile**.

Reaction	Projectile	Particle Input
$^{59}\text{Cu}(p, \gamma)^{60}\text{Zn}$	Proton	$1/2+$
$^{12}\text{C}(n, \gamma)^{13}\text{C}$	Neutron	$1/2+$
$^{16}\text{O}(\alpha, \gamma)^{20}\text{Ne}$	Alpha	$0+$

B. Single-Nucleon Transfer Reactions

Examples: (d, p) , (p, d) , $(^3\text{He}, d)$, (α, t)

In these reactions, a single nucleon is transferred between the projectile and the target. You must input the properties of the **transferred nucleon**, not the beam.

- **Target Input:** Spin/Parity of the target nucleus.
- **Particle Input:** Spin/Parity of the **transferred nucleon** (n or p).

Reaction Type	Example	Transferred Object	Particle Input
Pickup	$^{13}\text{C}(p, d)^{12}\text{C}$	Neutron	$1/2+$
Stripping	$^{40}\text{Ca}(^3\text{He}, d)^{41}\text{Sc}$	Proton	$1/2+$

Note on j-transfer: In transfer physics, we often discuss the transferred total angular momentum j (where $\vec{j} = \vec{\ell} + \vec{s}$). The code lists results by Channel Spin S . The set of allowed final J^π values is identical regardless of the coupling order.

C. Two-Nucleon Transfer Reactions

Examples: (p, t) , (t, p) , $(^3\text{He}, n)$, (α, d)

In these reactions, a pair of nucleons is transferred. The total spin of this pair depends on the coupling of the two nucleons.

1. Identical Nucleon Transfer (2n or 2p)

Reactions: (p, t) , (t, p) (2n transfer); $(^3\text{He}, n)$ (2p transfer)

For the transfer of two identical nucleons (2n or 2p) in the same shell model orbit, the **Pauli Exclusion Principle** dictates that their total wavefunction must be antisymmetric.

- If they are in a relative s-state (spatially symmetric, $L=0$), their **spin wavefunction must be antisymmetric**.
- An antisymmetric spin state for two fermions corresponds to a **Singlet State** ($S=0$).

- **ENSDF Policy:** For (p, t) , (t, p) , and $(^3\text{He}, n)$ reactions, it is standard to assume the transferred pair is in an **anti-parallel spin state** ($S=0$).
- **Particle Input:** 0+

2. Neutron-Proton Transfer (1n1p)

Reactions: (α, d) , (d, α) , $(^3\text{He}, p)$, $(p, ^3\text{He})$

For the transfer of a neutron-proton pair, the selection rules depend on the projectile/ejectile properties.

- **(α, d) and (d, α) :**
 - Both d and α have $T=0$. Thus, only **$T=0$** transfer is allowed.
 - Antisymmetry requires $S+T$ to be odd (for relative $L=0$). Since $T=0$, the transfer is restricted to **$S=1$** (spin triplet).
 - **Particle Input:** 1+
- **$(^3\text{He}, p)$ and $(p, ^3\text{He})$:**
 - Both particles have $S=1/2$, $T=1/2$.
 - Allowed transfers are **$(S=0, T=1)$** and **$(S=1, T=0)$** .
 - **Particle Input:**
 - For $S=0$ component: 0+
 - For $S=1$ component: 1+
 - *Note: Both components often contribute.*

D. Cluster Transfer Reactions

Examples: $(^6\text{Li}, d)$, $(^7\text{Li}, t)$

An alpha particle (or other cluster) is transferred.

- **Target Input:** Spin/Parity of the target nucleus.
- **Particle Input:** Spin/Parity of the **transferred cluster**.

Reaction	Transferred Cluster	Particle Input
$(^6\text{Li}, d)$	Alpha (^4He)	0+

E. Alpha Inelastic Scattering

Examples: (α, α')

In inelastic scattering, the target nucleus is excited to a higher energy state.

- **(α, α') :** The alpha particle has spin 0. Since the projectile spin cannot flip, the angular momentum transfer is purely orbital ($\vec{\ell}$). This selectively excites **Natural Parity** states ($\pi = (-1)^\ell$).
- **Particle Input:** 0+

◦ *Note: This treats the angular momentum transfer as a spin-0 boson transfer.*

F. Summary of All Particle Inputs

Reaction Category	Examples	Particle Input Rule	Input Value
Resonance / Capture	$(p,\gamma), (n,\gamma)$	Projectile Spin	1/2+
	(α,γ)	Projectile Spin	0+
Single-Nucleon	$(d,p), (p,d), (^3\text{He},d)$	Transferred Nucleon	1/2+
2-Nucleon (Identical)	$(p,t), (t,p), (^3\text{He},n)$	Anti-parallel Pair ($S=0$)	0+
2-Nucleon (Mixed)	$(\alpha,d), (d,\alpha)$	Triplet Pair ($S=1$)	1+
	$(^3\text{He},p), (p,^3\text{He})$	Mixed ($S=0$ & $S=1$)	0+ and 1+
Cluster Transfer	$(^6\text{Li},d)$	Transferred Cluster (α)	0+
Alpha Inelastic	(α,α')	Natural Parity Excitation	0+

3. Interpreting the Output

When you run the code:

```
python .github/scripts/angular_momentum_coupling.py 3/2- 1/2+
```

You receive output like:

```
Target: J=3/2 π=- | Particle: s=1/2 π=+
Channel Spins S: 1, 2: from |3/2 - 1/2| to 3/2 + 1/2
-----
L   Wave  Parity   s   Final Jπ
-----
0   s      -       1   1-
0   s      -       2   2-
-----
1   p      +       1   0+, 1+, 2+
1   p      +       2   1+, 2+, 3+
-----
```

- **L (Wave):** The orbital angular momentum (ℓ) between target and particle, ranging from 0 to 4 in the output (s, p, d, f, g waves).
- **Parity:** The parity of the final state, determined by $\pi_{\text{final}} = \pi_{\text{target}} \times \pi_{\text{particle}} \times (-1)^\ell$.

- **s (Channel Spin):** The intermediate coupling $S = \vec{J}_{\text{target}} + \vec{s}_{\text{particle}}$ before adding orbital angular momentum.
 - For $3/2^-$ target and $1/2^+$ particle: S ranges from $|3/2 - 1/2| = 1$ to $3/2 + 1/2 = 2$.
- **Final $J\pi$:** All allowed total angular momentum and parity combinations for each (L, S) pair, calculated from $|S - L| \leq J_{\text{final}} \leq S + L$.

Example Analysis: $^{59}\text{Cu}(p, \gamma)^{60}\text{Zn}$

- **Target:** ^{59}Cu ($3/2^-$)
- **Particle:** Proton ($1/2^+$)
- **Channel Spins:** $S = 1, 2$ (from $|3/2 - 1/2|$ to $3/2 + 1/2$)
- **Results:**
 - **s-wave capture** ($L=0$): Parity = $(-)(+) = -$
 - $S=1$: $J^\pi = 1^-$
 - $S=2$: $J^\pi = 2^-$
 - **p-wave capture** ($L=1$): Parity = $(-)(+) = +$
 - $S=1$: $J^\pi = 0^+, 1^+, 2^+$ (from $|1-1|$ to $1+1$)
 - $S=2$: $J^\pi = 1^+, 2^+, 3^+$ (from $|2-1|$ to $2+1$)
 - **d-wave capture** ($L=2$): Parity = $-$
 - $S=1$: $J^\pi = 1^-, 2^-, 3^-$
 - $S=2$: $J^\pi = 0^-, 1^-, 2^-, 3^-, 4^-$

4. Isospin Coupling

For reactions transferring identical nucleons (e.g., (p,t) , (t,p) , $(^3\text{He},n)$), isospin (T) provides additional constraints.

- **Identical Nucleon Pair ($2n/2p$):** In a relative $L=0$ state, the Pauli Principle requires $S=0$ (spin singlet) and $T=1$ (isospin triplet).
- **Selection Rule:** The final isospin T_f is determined by vector addition: $\vec{T}_f = \vec{T}_{\text{target}} + \vec{T}_{\text{transfer}}$. Since $T_{\text{transfer}} = 1$, the allowed final isospin values are $|T_{\text{target}} - 1| \leq T_f \leq T_{\text{target}} + 1$.

Note: This tool calculates angular momentum (J^π) only. Isospin selection rules must be applied separately.